

1 CP and ILP via Graph Burning

We will be working through some advanced work on CP and ILP modelling using the graph burning problem as a central theme. We will look at several papers on the topic, and compare different approaches looking at the development over time.

Here are links to the papers that we will be looking at, with a brief description of what we will discuss in each:

1.1 A survey on graph burning in general

<https://arxiv.org/abs/2009.10642>

- here we will talk about the definition of the graph burning problem, and mention the graph burning conjecture

1.2 CP and ILP formulations

<https://www.mdpi.com/2227-7390/10/15/2777>

- here we will look in fair detail at each of the CSP/ILP models for graph burning, and discuss how they work, how they differ and where they are similar.
- we will talk about the interaction with binary search
- we will look at the relative performances, and what the authors think about why these perform as they do

1.3 Parameterised complexity (this in less detail)

<https://arxiv.org/pdf/2007.08811>

- we will talk briefly about what parameterised algorithms are, and why and when they help us
- we will talk very briefly about why we would care about these algorithms

1.4 A greedy heuristic algorithm (less detail)

<https://arxiv.org/abs/2401.07577>

- we will talk about what a greedy algorithm is, what a heuristic algorithm is, and what they can and cannot do

- we will talk about the differences (and advantages and disadvantages) of heuristic, parameterised, and exact exponential algorithms
- we will very briefly mention the link to approximation algorithms, and what an approximation algorithm is

1.5 An exact row-generation algorithm (integer programming)

<https://arxiv.org/abs/2404.17080>

- here we will look at the updated ILP approach, and compare it briefly to the ones in the earlier paper
- we will look at the results here and talk about the comparison to other approaches